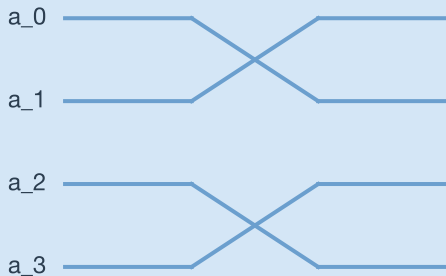


Algorithmic Contrast: NTT Butterfly vs. AMX Matrix TMVP

ML-KEM: $O(n \log n)$ NTT

Ring: $\mathbb{Z}_{3329}[x] / (x^{256} + 1)$



- Pre-transformed keys (domain amortization)
 - Pointwise multiply: 256 mul + Inv-NTT
 - Heavily optimized in ARM NEON SIMD

AMX: $O(n^2)$ TMVP Outer Product

Block-Toeplitz Convolution

32 × 32 Outer-Product Block

Sum along anti-diagonals:

$$c_k = \sum a_i \cdot b_{(k-i)}$$

(mac16 + EXTRH / VECINT flatten)

- Operates directly on coefficients
- Unamortized: 64 outer products for $n=256$
- Cannot beat $O(n \log n)$ NTT on NEON